Program overview

12-Sep-2023 11:14

Year	**2023/2024**
Organization	**Civil Engineering and Geosciences**
Education	**Minors Applied Earth Sciences**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
TA Minors 2023	**AES Minors 2023**		
TA-MI-077-23	**TA-MI-077-23 Research Project Applied Earth Sciences**		
TA3077	Research Project Applied Earth Sciences	30	
TA-MI-169-23	**TA-MI-169-23 Study abroad TA**		
TA-MI-192-23 Geo-resources for the Future			
AESB3110	Geo-resources 1.0: past and present	5	
AESB3111	Political economy of geo-resources	5	
AESB3113	Group project: future of geo-resources	15	
AESB3114	Geo-resources 2.0: towards the future	5	

Year	**2023/2024**
Organization	**Civil Engineering and Geosciences**
Education	**Minors Applied Earth Sciences**

TA Minors 2023

Year	**2023/2024**
Organization	**Civil Engineering and Geosciences**
Education	**Minors Applied Earth Sciences**

TA-MI-077-23	
Responsible Program Employee	Dr. K.H.A.A. Wolf
Minor Coordinator	Dr. K.H.A.A. Wolf
In association with the Faculty of	CITG
Administration by the Faculty of	CITG
Minor Title	Research Project Applied Earth Sciences
Contact for Students (Minor)	Dr. Karl-Heinz Wolf Associate Professor for Petrophysics TU Delft / Faculty of Civil Engineering and Geosciences Dept. of Geoscience & Engineering Section of Applied Geophysics & Petrophysics Stevinweg 1, 2628 CN, Room . 03.20 POB 5028, 2600 GA, Delft, The Netherlands. T +31 (0)15 278 6029 M +31 (0) 64 887 5958 E-Mail: K.H.A.A.Wolf@tudelft.nl
Intended for	All information can be found in the coursebase overview of TA3077
Gives access to	All information can be found in the coursebase overview of TA3077
Expected prior Knowledge	All information can be found in the coursebase overview of TA3077
Prerequisites Minor	All information can be found in the coursebase overview of TA3077
Minor Exit Qualifications	All information can be found in the coursebase overview of TA3077
Minor Coherence / Goal	All information can be found in the coursebase overview of TA3077
Minor Content 1	All information to be found in TA3077
Education Methods	All information can be found in the coursebase overview of TA3077
Minor Assessment	All information can be found in the coursebase overview of TA3077
Minor Remarks / Schedule	All information can be found in the coursebase overview of TA3077

TA3077	Research Project Applied Earth Sciences	30

Responsible Instructor	Dr. K.H.A.A. Wolf
Contact Hours / Week x/x/x/x	
Education Period	1 2 3 4
Start Education	1 2 3
Exam Period	none
Course Language	Dutch English
Required for	BSc 2nd year
Expected prior knowledge	1. You followed the basic courses for a BSc at Delft University, or elswhere, with general courses such as intros in maths, physics, chemistry, statistics, mechanics, etc.. 2. In addition, you may have followed or will attend introductions in geoscience and specific introductions associated to a topic of your choice. 3. You finished your 1st year and got at least 30 ECT of your 2nd year program (P+30)
Parts	-
Course Contents	The projects are connected to an educational program within the Department of Geoscience and Engineering. When you are following a BSc-program in another Department, Faculty or University, the projects will be linked to YOUR BSc-curriculum and OUR Geoscience & Engineering program. Parts of the research minor can be done outside the faculty and Delft University. In addition, topics can also be related to a bridging minor in Geo-engineering organized by Prof.Dr.M.A. Hicks and Dr.R.D.J.M. Ngan-Tillard. In a meeting with Dr.Karl-Heinz Wolf you discuss your background and research wish. Then we try to connect it to one of the available research topics or a proposal for a new topic.
Study Goals	You are able to perform individually (fundamental, strategic and/or applied) GEOSCIENCE AND ENGINEERING CURRICULUM RELATED TOPICS. The guidelines are the "eindtermen BSc Aardwetenschappen" as composed by the "Kamer Aardwetenschappen (VSNU)".
Education Method	1. Develop a project plan. 2. Do the research. 3. Prepare a report for review and judgement. 4. Give an oral presentation.
Assessment	A report and oral presentation. Before you get your mark, you hand in: 1. A pdf of the presentation. 2. A pdf of the final report. 3. A time sheet, how you spent your time within your research up to a total of 30EC, or 840 hours. 4. From your supervisor(s) we receive a mark and a short motivation why you received this mark. Note that the compulsory TU-Delft supervisor and second committee member are licenced to educate. More assessors (PhD, Postdoc, supervisor from industry, etc.) are very welcome. Note that you finalize a BSc-curriculum topic in the Geoscienec and Engineering Department. If you are from outside our department, we strongly advised you to have one reviewer from your own BSc-curriculum present in your committee. 1. Follow the guidelines explained in the section "assessment", here above. If that is not possible, use the judgement form (Rubrics excel)from the bachelor end project (AESB3400). 2. The compulsory TU-Delft supervisor and second committee member are licenced to educate and are working in two different sections. Again, other assessors (PhD, Postdoc, supervisor from industry, etc.) are welcome, however not allowed to give the official mark. Again: Note that you finalize a BSc-curriculum topic for Applied Earth Science.
Special Information	Dr. K-H.A.A. Wolf ____________________________ Associate Professor in Petrophysics TU Delft / Department of Geoscience & Engineering Stevinweg 1, 2628 CN, Delft The Netherlands. Office: +31 (0)15 278 6029 Mobile: +31 (0)648 875 958 E-mail: k.h.a.a.wolf@tudelft.nl
Contact	For the research minor topics in Geoscience and Engineering, please contact Dr. Karl-Heinz Wolf, Associate Professor in Petrophysics. Room 03.20, T: +31 (0)15 278 6029, E: k.h.a.a.wolf@tudelft.nl For the transition minor, please contact: 1. Prof.Dr.M.A. Hicks, E: M.A.Hicks@tudelft.nl 2. Dr.Ir.D.J.M.Ngan-Tillard, E: D.J.M.Ngan-Tillard@tudelft.nl
Expected prior Knowledge	P + 30 ECTS and attended all lectures and practicals, which are relevant for your research topic.
Academic Skills	1. You followed the basic courses for a BSc at Delft University or elswhere with general courses such as intros in maths, physics, chemistry, statistics, mechanics, etc.. 2. In addition, you may have followed or will attend introductions in geoscience and specific introductions associated to a topic of your choice. 3. You finished your propedeuse and got at least 30 ECT of your 2nd year program (P+30)
Literature & Study Materials	It depends on your topic. You define at the start, in consultation with your supervisor(s), what you need to know and what you need to learn, to perform in order to finalize a research minor.
Judgement	1. Follow the guidelines explained in the section "assessment", here above. If that is not possible, use the judgement form (Rubrics excel)from the bachelor end project.

	2. The compulsory TU-Delft supervisor and second committee member are licenced to educate and are working in two different sections. Additionally, other assessors (PhD, Postdoc, supervisor from industry, etc.) are welcome, however, they are not allowed to give an official mark. Note that you finalize a BSc-curriculum topic for Applied Earth Science. Therefore, it is strongly advised to have one reviewer from your Department present in your committee. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/.
Permitted Materials during Exam	-
Collegerama	No

TA-MI-169-23	
Responsible Program Employee	N.H. Can
Contact for students	Exchange-citg@tudelft.nl, room HG2.73
In association with the Faculty of	CITG
Prerequisites	a. students in their second year can apply for a minor abroad (in their third year) if they have a positive BSA (a postponed BSA doesnt qualify) upon submitting their exchange nomination package; b.) students who apply for a minor study abroad in their 3rd year (or later) of registration, should have received at least a positive BSA and gained in total min. 75 EC in the BSc, upon submitting their exchange nomination package; c. to qualify for a minor abroad, students need to comply with the concerned administrative procedure, see the exchange manual of the year concerned. Further depending on the course package chosen.
Program Goals	BSc-student is required to follow a consistent package of courses within a specific theme worth 30 EC. The course package may broaden or deepen student's knowledge and may not overlap the BSc major and your future MSc programme. The level of the course package should lead to 3rd year BSc level. Student studies 1 semester at a foreign partner university. Besides student experiences the way of living and studying in another cultural environment and expands his/her foreign language skills. All conditions for the free minor are to be found on the following website: https://www.tudelft.nl/en/student/ceg-student-portal/education/bachelor/self-composed-minor-information The package of courses is composed by the student him-/herself and needs to be officially approved by the minor coordinator CIE/AES, before the study period abroad. This is part of the exchange nomination process. Please mind following (BSc-students): it is the responsibility of the student him-/herself to gain 30 EC for the minor abroad. If student does not succeed, student needs to take care of a suitable supplement in Holland to complete the Free Minor Study Abroad. This may lead to study delay. Therefore, it is strongly advised to participate in the Free Minor Study Abroad only if student is studying nominally. MSc-student is required to follow courses worth min. 20 EC after consultation with the MSc coordinator. Grades obtained at the partner university will be converted to pass/fail. On the diploma supplement the original grades will be visible.
Exam requirements	BSc-student: the package of courses is composed by the student him-/herself and needs to be officially approved by the free minor commission CIE/AES, before the study period abroad. This is part of the exchange nomination process. Student fills in the form 'Application Approval Free Minor' and asks a lecturer, specialized in the theme of the minor, to sign this form as well showing his/her support as far as the consistency of the course package is concerned. After the study period abroad, student is required to hand in the official transcript of grades with a written request to the Board of Examiners CeG as soon as possible. MSc-student: Check article 3C of the TER Annex of your study and starting year. This article describes how to go about electives in your study programme. Consult your track coordinator for approval of your proposed courses. The MSc coordinator has to approve your MSc curriculum by signing the AES/CIE-2 form.
Intended for	AES-students 5th semester BSc program,MSc-AES students. Please mind following (BSc-students): it is the responsibility of the student him-/herself to gain 30 EC for the minor abroad. If student does not succeed, student needs to take care of a suitable supplement in Holland to complete the Free Minor Study Abroad. This may lead to study delay. Therefore, it is strongly advised to participate in the Free Minor Study Abroad only if student is studying nominally.
Maximum number of participants	See the most updated overview with exchange places CeG available on Brightspace (Student and International Office CeG > Exchange).
Remarks Maximum number of participants	depending on the number of exchange places available, which may differ per university and per study year.
Minimum number of participants	0
Minor Remarks / Schedule	As from the beginning of the study year the Student and International Office CeG (SIO) will organize exchange information meetings. Please keep yourself informed of the data and other news by reading Brightspace and/or the announcements on the publication boards of SIO. All available exchange places CeG for both semester 1 and 2 and for both BSc and MSc students CeG will be set out through 2 nomination rounds. Please enroll Brightspace to keep yourself updated, and visit the exchange information meetings!The exchange manual which can be found on Brightspace is part of the exchange procedure so read this manual carefully! Please mind following: it is the responsibility of the student him-/herself to gain 30 EC for the minor abroad. If student does not succeed, student needs to take care of a suitable supplement in Holland to complete the Free Minor Study Abroad. This may lead to study delay. Therefore, it is strongly advised to participate in the Free Minor Study Abroad only if student is studying nominally. At the foreign partner universities different application deadlines and -conditions are applicable. Please keep in mind your own specific time planning. It is often not allowed to follow courses other than CeG at a foreign university. If you wish to follow courses in a completely different study area to broaden your knowledge, you need to contact the partner university yourself and ask for written permission for this. This may affect scholarship options. Students who spend an exchange period abroad, are required to send SIO digital experience reports during and after the exchange stay, and to act like a TU Delft ambassador upon request of SIO. The experience reports will be published on the faculty study abroad website and/or Brightspace. These reports are different from the ones you need to write for your (scholarship) application! After return, students also commit themselves to actively particpate in the buddyprogram for incoming international students Finally it is the full responsibility of the student to start the exchange- and application procedure properly and in time and to meet the prerequisites and language requirements from the partner university in time as well. If we notice that during any moment of the exchange procedure (before and after nomination) you do not show a proper and flexible attitude towards the procedures, the Student and International Office and/or the host university keep the right of cancelling your nomination and to not support your exchange study any longer. Student needs to keep him-/herself updated of the latest exchange information, provided by exchange information meetings, the exchange manual and other communication channels like www.studyabroad.tudelft.nl, the faculty study abroad website, Brightspace.

	No correspondence is possible about the outcome of f.e. the exchange nomination process. No withdrawal of the nomination (package) is possible. Links: www.studyabroad.tudelft.nl www.studyabroad.citg.tudelft.nl www.brightspace.tudelft.nl > Student and International Office CeG > Exchange

Year	**2023/2024**
Organization	**Civil Engineering and Geosciences**
Education	**Minors Applied Earth Sciences**

TA-MI-192-23 Geo-resources for the Future	
Responsible Program Employee	Dr. A.M.H. Pluymakers

AESB3110	Geo-resources 1.0: past and present	5
Responsible Instructor	Prof.dr. S. Geiger	
Instructor	Dr. A.M.H. Pluymakers	
Contact Hours / Week x/x/x/x	first 5 weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is primarily part of the Minor 'Geo-resources for the Future: Social, Economic, Environmental and Political Challenges'. This module gives a geological and engineering introduction to geo-resources. Some economic, political and social aspects will inherently be included, but is not the primary focus. The course has a number of aspects: - Self study: Geo-MOOC to understand basic relevant aspects of geology. - Lectures: a selection of mini-lecture series on a specific geo-resources will be given. The focus of each mini-series will be 20% technical, 20% project development and 60% on (global) trends, social aspects, political aspects, environmental aspects, etc.. The focus will be on current resources and how they have been utilised in the past and today. The mini-series topics are (may change): (i) Mining and Minerals, (ii) Oil and Gas, (iii) Waste Storage, and (iv) Energy. - Forum activities: the 'forum' contains joint activities across the minor. Summative assessment will take place in the forum activities.	
Study Goals	After completion of this module the student will be able to: 1. Explain basic features of the subsurface, including the origin and basic properties. 2. Identify the variety of resources in the subsurface, including location and geological features. 3. Give examples of how geo-resource projects have been developed. 4. Distinguish and describe how different uses of the surface and sub-surface can interact and interfere with each other. 5. Explore how economic, political and societal aspects can influence geo-resource development.	
Education Method	Lectures, online self-study (MOOC), Wiki development, 'Forum' activities	
Course Relations	Students are expected to follow the courses AESB3111 and AESB3112, which run in the same quarter.	
Assessment	Summative assessment is via forum activities. 60% Wiki development 20% Essay 20% Panel discussion and policy advice Participation in non-graded activities is compulsory	
Elective	Yes	
Tags	Analysis Broad Energy & Industry Geo Engineering Geology Group work Information & Communication Integrated Project planning / management Projects Sustainability	
Expected prior Knowledge	This course does not require prior knowledge for entry, but where deficiencies are identified, e.g. in geology, students are required as part of the course to repair knowledge, e.g. via the online study material.	
Academic Skills	Knowledge assimilation, critical thinking, team work	
Literature & Study Materials	To repair geological knowledge the Geo-MOOC is used. For the rest of the course there is no fixed study material. Students are expected to delve into available literature to form a wider understanding.	
Judgement	60% Wiki development 20% Essay 20% Panel discussion and policy advice Participation in non-graded activities is compulsory For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

AESB3111	Political economy of geo-resources	5
Responsible Instructor	Dr. A.F. Correlje	
Co-responsible for assignments	G.O. Ndubuisi	
Contact Hours / Week x/x/x/x	2 hours during the entire period and during week 5 to 7 one day	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The module discusses the theory and practice of selecting and implementing the adequate forms of economic coordination in the design, construction and exploitation of infrastructures and natural resources. Examples are the choice between public, private or collective ownership, contractual arrangements, tariff structures, forms of cooperation, public intervention and oversight and regulation. We apply notions as developed in the neo-classical and institutional economic theory.	
Study Goals	The objectives of the module Economics of Infrastructures are to develop the ability: to reflect on the most important issues in in economics of infrastructures, with a focus on grid bound systems for energy and water supply, communication and transport. to describe the differences between neo-classical, new institutional and the original institutional economics, referring to the assumptions underlying these economic schools, the specific theories and the relevant methods of analysis. to judge, referring to these differences, what specific infrastructural issues should be analysed and understood with which economics approaches and concepts. to identify, describe and interpret these issues, applying the paradigms and concepts of the different schools. to critically reflect on public policy and firms strategies in the infrastructures sector. to identify the nature of the problems in public policy and firms management in the provision of infrastructural services, considering societal and technological developments. To communicate about such issues, applying the appropriate economic terminology.	
Education Method	Lectures * Lectures made available on Collegerama. Assignments, provided on BS. Discussion of the assignments during the tutorials.	
Assessment	- Participation in tutorials, evaluation of assignments - Exam with open questions In case of unforeseen circumstances or measures resulting from COVID-19 or otherwise, the prescribed assessment will be modified as follows: timed take-home exam (open book). This contains several open-ended essay questions in a restricted time period, comparable to the on-campus exam. A working internet connection is the responsibility of the students.	
Expected prior Knowledge	None	
Academic Skills	Learning to judge what specific infrastructural issues should be analysed and understood with which economics approaches and concepts.	
Literature & Study Materials	Material will be made available on BS or links will be provided.	
Judgement	Online open book Exam In case of unforeseen circumstances or measures resulting from COVID-19 otherwise, the prescribed assessment will be modified as follows: timed take-home exam (open book). This contains several open-ended essay questions in a restricted time period, comparable to the on-campus exam. A working internet connection is the responsibility of the students. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Dictionary	
Collegerama	No	

AESB3113	Group project: future of geo-resources	15
Responsible Instructor	Dr. A.M.H. Pluymakers	
Contact Hours / Week x/x/x/x	0/1 whole day and a half/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This is the group project of the minor 'Geo-resources for the future: social, economic, environmental and political challenges'. Students will work in multidisciplinary teams of 4-5 members on a specific case study. The aim is to deepen and combine the skills and knowledge acquired in the Q1 courses of the minor. A list of topics (with supervisors) will be made available in quarter 1. Students may propose their own topic, to be discussed and approved by the minor coordinator. Each topic falls within one of the following themes (may change): 1. Energy transition 2. Mining / materials 3. Use of underground space Students will be able to give their preferences (first, second, third choice), based on which the coordinator will assign the topics. Thereby a balanced mix of students from different backgrounds over the groups is favoured. During the group project, students will experience what it is like to work in a project setting, where they are responsible for defining the project aims, the planning, communicating with each other and external parties (supervisors and peers), reporting, etcetera. The project has a kick-off in week 1, with a first meeting in week 2 for information sharing. In weeks 2, 4, 6 and 8 the groups working in the same theme present their plans and progress to each other and their supervisors. In week 5 a mid-term review will be organised with presentations, self- and peer-assessment and evaluation meetings (with team and individual). Similarly, the final review will be held in week 10 with a symposium day open to the public.	
Study Goals	After completing the group project students will be able to: - provide a valuable contribution in the search for solutions related to geo-resources for future society, taking into account technological, environmental, geo-political and economical aspects in an international context; - analyse the institutional, legal and economical playing field; - assess the different aspects, their interactions and impacts; - reflect on and communicate the aforementioned analysis and assessment; - analyse the role and influence of industry and institutions in an international context; - present the technological, institutional and economical value chain for a given geo-resource including historical knowledge: awareness of how the value chains have developed; - analyse the impacts of (climate) policies, consequences for resource exploration/development and subsidies; - transfer a balanced overview of the energy/geo-resource landscape; - act in a socially responsible way in the geo-resource industry or relevant institutions; - effectively communicate their plans, progress, feedback to team members both written and orally; - work in a multi-disciplinary team: showing a professional attitude, taking initiatives, managing their work.	
Education Method	group project	
Assessment	- peer feedback (formative only) - 30% presentations + discussion (group and individual grade) - 50% group report (group and individual grade) - 20% skills to work in multi-disciplinary team at expert level (individual grade based on interaction with supervisors, ability to show and defend own contribution). The work is graded individually. The final grade for the group project will be a combination of the group grade and the individual grade. More detailed information on the grading including the grading criteria is made available at the start of the project. Grading takes place around the time of the mid-term review and in the week after the final review. Students will only be awarded a sufficient final grade for the project if: 1. The whole project has been completed and, 2. The group grade is 5.0 or higher, 3. The individual grade is 5.0 or higher, 4. The final grade is 5.8 or higher. The final grade will be rounded to the nearest half grade.	
Remarks	Student attendance and commitment is absolutely vital to the attainment of the best possible learning outcomes of the group project. The project requires a full time commitment of the student. It is recommended not to undertake any other study activity beside the project. Insufficient participation of an individual may lead to expulsion from the project.	
Tags	Analysis Broad Challenging Economics Energy Finance Geo Engineering Geology Group Dynamics/Project Organisation Group work Information & Communication Integrated Intensive Involved Policy Analysis Process Project Project planning / management Research Methods Sustainability Technology Underground	
Expected prior Knowledge	Students must have completed the courses AESB3110, AESB3111, AESB3112.	

Academic Skills	Knowledge assimilation, critical thinking, analysis, team work, communication, project management.
Literature & Study Materials	Literature and online materials related to the project must be searched for by the students.
Judgement	The work is graded individually. The final grade for the group project will be a combination of the group grade and the individual grade. More detailed information on the grading including the grading criteria is made available at the start of the project. Grading takes place around the time of the mid-term review and in the week after the final review. Students will only be awarded a sufficient final grade for the project if: 1. The whole project has been completed and, 2. The group grade is 5.0 or higher, 3. The individual grade is 5.0 or higher, 4. The final grade is 5.8 or higher. The final grade will be rounded to the nearest half grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Not applicable.
Collegerama	No

AESB3114	Geo-resources 2.0: towards the future	5
Responsible Instructor	Dr. E.G.M. Kleijn	
Instructor	Drs. L.C. van Geuns	
Contact Hours / Week x/x/x/x	weeks 6-9	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is a part of the minor 'Geo-resources for the future: Social, Economic, Environmental and Political Challenges'. It builds upon the other modules of this minor, being 'Geo-resources 1.0: past and present' (AESB3110) and 'Political Economy of geo-resources' (AESB311). This course is designed around five themes: 1. The Global Resource Nexus: linkages between (supply and demand) of the different geo-resources, scarcity of those resources, and climate change. 2. Global consequences of the resource nexus, focusing on geo-politics, security and strategy. 3. Licence to operate, considering corporate responsibility and local acceptability for specific geo-resources. 4. Energy transition and climate change. 5. Circular economy, covering aspects like urban mining, recycling and certification.	
Study Goals	After completion of this module the student is able: - to set up a scenario analysis with relation to future resource supply and demand; - to describe the dilemmas related to an increased energy demand and the need to decrease CO2 emissions; - to analyse the interactions between politics, policies, regulations related to geo-resources exploitation; - to understand the environmental impacts of mining, including linkages to other resources like water, energy, land and biodiversity; - to understand the concepts of Circular Economy and circular business models as a means to reduce demand for primary resources; - to analyse the challenges in this field related to various interests; - to evaluate characteristics of societal aspects of resource exploitation; - to distinguish and put into context the role of various stakeholders with respect to corporate social responsibility (ethical reflection); - to contribute to policy development and firm strategies taking into account uncertainties in the process of resource exploitation.	
Education Method	Lectures and online video's, workshops, debate, game, news forum.	
Assessment	Assessment via 'forum' activities: wiki development, essay, debate, game, news forum 40% Essay 60% Assignments Participation in non-graded forum activities is compulsory.	
Tags	Analysis Broad Challenging Economics Energy Finance Policy Analysis Small groups Sustainability	
Expected prior Knowledge	Students must follow (or have followed) the courses AESB3110 and AESB3111, which run in the same quarter.	
Academic Skills	Knowledge assimilation, critical thinking, communication, team work	
Literature & Study Materials	Online materials, topical articles and reports will provided via Brightspac	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable.	
Collegerama	No	

Dr. A.F. Correlje

Unit	Techniek, Bestuur & Management
Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 87924

Prof.dr. S. Geiger

Department	Applied Geology
Telephone	+31 15 27 82203
Room	23.HG 3.22

Drs. L.C. van Geuns

Department	Geo-engineering

Dr. E.G.M. Kleijn

Unit	Techniek, Bestuur & Management
Department	Onderwijs en Studentenzaken

G.O. Ndubuisi

Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 88895

Dr. A.M.H. Pluymakers

Unit	Civiele Techniek & Geowetensch
Department	Applied Geophysics and Petrophysics
Telephone	+31 15 27 87599
Room	23.HG 3.30

Dr. K.H.A.A. Wolf

Unit	Civiele Techniek & Geowetensch
Department	Applied Geophysics and Petrophysics
Telephone	+31 15 27 86029
Room	23.HG 3.20

ontbreekt

N.H. Can